

$$\int \frac{1}{e^z}$$

$$\text{since } e^z = 1 + z + \frac{1}{2!}z^2 + \frac{1}{3!}z^3 + \dots,$$

$$e^{\frac{1}{z}} = 1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots \quad (|z| > 0)$$

$$\text{so, } \text{res}(e^{\frac{1}{z}}, 0) = 1, \text{ therefore } \int_{|z|=1} e^{\frac{1}{z}} dz = 2\pi i.$$

Thm. Residue Theorem.

Let f be an analytic inside and on a simple closed contour C except iso. sing. $z_1, \dots, z_n$.

Then
$$\int_C f(z) dz = 2\pi i \cdot \sum_{k=1}^n \text{Res}(f, z_k).$$

Lemma. If f has a pole of order m at z_0 , then

$$\text{Res}(f, z_0) = \frac{1}{(m-1)!} \cdot \lim_{z \rightarrow z_0} \left((z - z_0)^m \cdot f(z) \right)^{(m-1)}$$

Example.

Evaluate $\int_{-\infty}^{\infty} \frac{1}{x^2+1} dx = \pi.$

Let $f(z) = \frac{1}{z^2+1}$. It has two isolated singular $z = \pm i$.

And, $\frac{1}{z^2+1} = \frac{1}{z-i} \cdot \frac{1}{z+i}$, so it has a simple pole, so

$$\text{Res}(f, i) = \lim_{z \rightarrow i} (z-i) \cdot \frac{1}{z^2+1} = \frac{1}{2i}$$

and similarly $\text{Res}(f, -i) = -\frac{1}{2i}$.

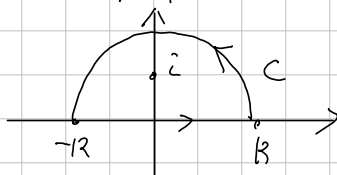
Now, for half circle C containing i ,

$$\int_C \frac{1}{z^2+1} dz = \int_{-R}^R \frac{1}{x^2+1} dx + \int_0^\pi \frac{iRe^{i\theta}}{R^2e^{2i\theta}+1} d\theta$$

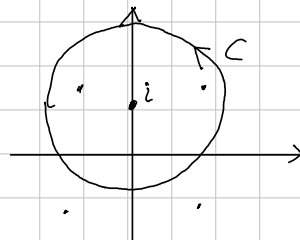
$$\text{And } \left| \int_0^\pi \frac{iRe^{i\theta}}{R^2e^{2i\theta}+1} d\theta \right| \leq \int_0^\pi \left| \frac{iRe^{i\theta}}{R^2e^{2i\theta}+1} \right| d\theta \leq \int_0^\pi \frac{R}{R^2-1} d\theta = \frac{R\pi}{R^2-1} \rightarrow 0 \text{ as } R \rightarrow \infty.$$

($|R^2e^{2i\theta}+1| \geq |R^2e^{2i\theta}|-1$) so,

$$\int_{-\infty}^{\infty} \frac{1}{x^2+1} dx = \int_C \frac{1}{z^2+1} dz = 2\pi i \cdot \frac{1}{2i} = \pi.$$



Evaluate $\int_C \frac{1}{z^4+4} dz$ where $C = \{z \in \mathbb{C} \mid |z-i|=2\}$.



Proof. Since z^4+4 has four roots

$$\sqrt{2} \cdot \exp\left(\frac{1}{4} \cdot \pi i\right), \sqrt{2} \exp\left(\frac{3}{4} \cdot \pi i\right), \sqrt{2} \exp\left(\frac{5}{4} \pi i\right), \sqrt{2} \exp\left(\frac{7}{4} \pi i\right),$$

so, by the Residue Theorem,

$$\begin{aligned} \operatorname{Res}(f, a_0) + \operatorname{Res}(f, a_1) &= \lim_{z \rightarrow a_0} \frac{z-a_0}{z^4+4} + \lim_{z \rightarrow a_1} \frac{z-a_1}{z^4+4} \\ &= \frac{1}{(a_0-a_1)(a_0-a_2)(a_0-a_3)} + \frac{1}{(a_1-a_0)(a_1-a_2)(a_1-a_3)} \end{aligned}$$

$$\text{Since } a_0 = \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}} \right) = 1+i$$

$$a_1 = \sqrt{2} \left(-\frac{1}{\sqrt{2}} + i \cdot \frac{1}{\sqrt{2}} \right) = -1+i$$

$$a_2 = -1-i$$

$$a_3 = 1-i,$$

$$\operatorname{Res}(f, a_0) + \operatorname{Res}(f, a_1) = \frac{1}{2 \cdot (2+2i)(2i)} + \frac{1}{(-2) \cdot (2i) \cdot (-2+2i)}$$

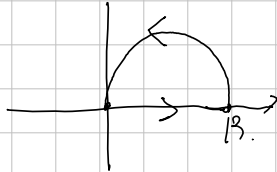
$$\begin{aligned} \text{so } \int_C \frac{1}{z^4+4} dz &= 2\pi i \cdot (\operatorname{Res}(f, a_0) + \operatorname{Res}(f, a_1)) \\ &= \frac{\pi}{4+4i} + \frac{\pi}{4-4i} = \frac{8\pi}{32} = \frac{\pi}{4}. \end{aligned}$$

Evaluate $\int_0^{\infty} \frac{1}{1+x^6} dx$.

Let $f(z) = \frac{1}{1+z^6}$. Then, since $z^6 = -1$ has six roots as

$$a_k = \exp\left(2\pi i \cdot \frac{k}{6} + \frac{\pi i}{6}\right) \quad (k=0,1,2,\dots,5)$$

Let a contour C_R as



As $R \rightarrow \infty$, C_R contains only a_0 , so by the Residue Theorem,

$$\int_{C_R} f(z) dz = 2\pi i \cdot \text{Res}(f; a_0) = \lim_{z \rightarrow a_0} (z - a_0) \cdot f(z) = 1 / \prod_{i=1}^5 (a_0 - a_i)$$

Meanwhile,

$$\int_{C_R} f(z) dz = \int_0^R \frac{1}{1+x^6} dx + \int_0^{\pi} \frac{iR e^{i\theta}}{1 + (R e^{i\theta} + \frac{R}{2})^6} d\theta$$

$$Z(t) = R \cdot e^{i\theta} + \frac{R}{2}$$

Evaluate

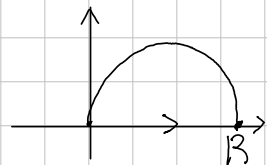
$$\int_0^{\infty} \frac{1}{1+x^n} dx \quad \text{for } n=2, 3, \dots$$

Proof: Let $f(z) = \frac{1}{1+z^n}$.

Since $z^n = -1$ has roots as

$$a_k = \exp\left(\frac{1}{n} \cdot (2k\pi i + \pi i)\right) \quad (k=0, 1, \dots, n-1)$$

Construct a contour C_R as



$$0 \leq \frac{\pi i}{n} \cdot (2k+1) \leq \frac{\pi i}{2}$$

$$0 \leq 4k+2 \leq n$$

As $R \rightarrow \infty$, the contour contains inside

$$a_1, a_2, \dots, a_m, \quad \text{where } m = \max \{i \in \{0, \dots, n-1\} \mid 4i+2 \leq n\}.$$

Now, by the Residue Theorem,

$$\int_{C_R} \frac{1}{1+z^n} dz = 2\pi i \cdot \sum$$

